

Sport and Health: Predicting Health Conditions from Fitness and Lifestyle Data

Group Members :

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Project Summary

This project investigates the relationship between physical activity and health outcomes, using a public health and fitness dataset of over 680,000 recorded exercise sessions from 3,000 participants over a year.

The Problem ? Despite widespread awareness that exercise is beneficial, the extent to which everyday lifestyle factors (activity type, duration, intensity, sleep, stress, hydration, and daily steps) relate to the presence of common health conditions (hypertension, diabetes, and asthma) is rarely shown in an accessible way. Our project aims to demonstrate, with data, why staying active is associated with better health and a lower likelihood of chronic conditions.

Solution:

After exploring and cleaning the dataset, we will use relevant information and features (such as BMI, activity intensity score, sleep quality) and train and compare the data with different classification models studied in class, such as logistic regression, decision tree, ... to predict whether an individual is healthy or likely to present a given health condition based on their lifestyle and fitness indicators.

Model performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices, with particular attention for class imbalance, since the majority of records correspond to healthy individuals.

Expected results:

We expect to find that higher activity levels, better sleep quality, and lower stress are associated with reduced likelihood of diseases like hypertension and diabetes while acknowledging the dataset's limitations (synthetic or self-reported nature, and likely modest predictive performance given the random-like relationships between lifestyle and health labels).

Beyond the predictive task, the project will try to provide solid, data-driven evidence supporting the importance of regular physical activity for long-term health, while still being honest and cautious, discussing the model's limitations and the responsible use of health-related predictions.